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COURSE NAME: FUNDAMENTAL OF MATHEMATICS.

[SET :01]

Q.No. 1:-

Answer(1):- Given set $\{1,2,3\}$, where $R = (1,1),(2,2),(3,3),(1,2),(2,3)$. We determine whether this relation is reflexive, symmetric, and transitive.

1). Reflexivity:

A relation is reflexive if, for every element x in the set, the pair (x,x) is included in R .

If the relation is reflexive then,

$(a,a) \in R$.

$(1,1),(2,2),(3,3) \in R$.

So, this relation is reflexive.

2). Symmetric:

A relation is symmetric if, whenever a pair (a,b) is included in R , and the reverse pair of this set is also included in the R .

$(a,b) \in R$

$(b,a) \in R$.

In this set we can see $(1,2) \in R$ but $(2,1)$ not belongs to R .

So, this relation is not symmetric.

3). Transitive:

A relation is transitive if, whenever (a,b) and (b,c) are in R , the pair (a,c) is also included in R .

$(a,b) \in R$

$(b,c) \in R$

$(a,c) \in R$

$(1,2) \in R$ $(2,3)$ is also belongs to R but $(1,3)$ is not belongs to R .

So, this relation is not transitive.

- **Summary:**
 - The relation R is reflexive because every element in the set has the pair.
 - The relation R is not symmetric because there reverse pair is not included in the relation R.
 - The relation R is not transitive because (1,2),(2,3) are included in the relation R but (1,3) is not included in the relation R.

Q No. 2:-

Answer(2):- (a). $f(x) = x^2$ and $g(x) = 2x+1$.

We need to find the composite function $f\{g(x)\}$ where $f(x) = x^2$ and $g(x) = 2x+1$.

$$f\{g(x)\} = (2x+1)^2$$

$$\text{then, } (2x+1)^2 = 4x^2 + 4x + 1$$

therefore, the composite function is:

$$f\{g(x)\} = 4x^2 + 4x + 1.$$

$$(b). f(x) = x+1 \text{ and } g(x) = x^3 + \sin x$$

In this part we have given $f(x) = x+1$ and $g(x) = x^3 + \sin x$

$$f\{g(x)\} = (x^3 + \sin x) + 1.$$

Simplifying the expression:

$$f\{g(x)\} = (x^3 + \sin x) + 1.$$

Thus, the composite function is:

$$f\{g(x)\} = x^3 + \sin x + 1$$

- So, final answer is:
 - For part (a), the composite function is $f\{g(x)\} = 4x^2 + 4x + 1$.
 - For part (b), the composite function is $f\{g(x)\} = x^3 + \sin x + 1$.

Q No. 3:-

Answer(3):-

$$(i). \lim_{n \rightarrow \infty} \frac{3n^2 + 4n + 7}{2 + 3n + 4n^2}$$

Now we divide the numerator & denominator by n^2 , the highest power of n :

$$\frac{3n^2+4n+7}{2+3n+4n^2} = \frac{\frac{3n^2}{n^2} + \frac{4n}{n^2} + \frac{7}{n^2}}{\frac{2}{n^2} + \frac{3n}{n^2} + \frac{4n^2}{n^2}}$$

$$\text{Or, } \frac{3 + \frac{4}{n} + \frac{7}{n^2}}{\frac{2}{n^2} + \frac{3}{n} + 4}$$

$n \rightarrow \infty$, terms like $\frac{4}{n^2}$, $\frac{7}{n^2}$, $\frac{2}{n^2}$, and $\frac{3}{n}$ then we saw,

$$\frac{3 + \frac{4}{\infty} + \frac{7}{\infty}}{\frac{2}{\infty} + \frac{3}{\infty} + 4} = \frac{3+0+0}{0+0+4}$$

Thus, the value of the limit is: $\frac{3}{4}$ ans.

$$(ii). \lim_{n \rightarrow \infty} \frac{x^2-3x+2}{x-2}$$

Simplify the numerator (x^3-3x+2)

$$(x^2-3x+2)$$

$$\text{Or, } (x^2-x-2x+2)$$

$$\text{Or, } x(x-1) - 2(x-1)$$

$$\text{Or, } (x-1)(x-2)$$

$$\text{Now, } \frac{(x-1)(x-2)}{(x-2)}$$

$$\text{So, } (x-1)$$

Now, find the limit as $x \rightarrow 2$;

$$\lim_{n \rightarrow 2} (x-1) = (2-1) = 1.$$

So, the value of the limit is: 1.

[SET:02]

Q No. 4:-

Answer(4):- We have given this equation:

$$y = \frac{\cos x + \sin x}{\sin x - \cos x}$$

we can solve this equation according to "quotient rule", which is given as follows:

$$\frac{(\sin x - \cos x) \cdot d\left(\frac{\cos x + \sin x}{dx}\right) - (\cos x + \sin x) \cdot d\left(\frac{\sin x - \cos x}{dx}\right)}{(\sin x - \cos x)^2}$$

$$\text{Or, } \{(\sin x - \cos x) \cdot (-\sin x + \cos x)\} - \{(\cos x + \sin x) \cdot (\cos x + \sin x)\} / (\sin x - \cos x)^2$$

$$\text{Or, } \frac{-\{(sinx-cosx).(sinx-cosx)-\{(cosx+sinx)^2\}}{(sinx-cosx)^2}$$

$$\text{Or, } \frac{-\{(sinx-cosx)^2\}-\{(cosx+sinx)^2\}}{(sinx-cosx)^2}$$

$$\text{Or, } \frac{-\{sin^2x+cos^2x-2sinx.cosx\}-\{cos^2x+sin^2x+2sinxcosx\}}{(sin^2x+cos^2x-2sinx.cosx)}$$

$$\text{Or, } \frac{-2sin^2x-2cos^2x}{1-2sinx.cosx}$$

$$\text{Or, } \frac{-2(sin^2x+cos^2x)}{1-2sinx.cosx}$$

$$\text{Or, } \frac{-2(1)}{1-sin2x}$$

$$\text{Or, } \frac{-2}{1-sin2x}$$

$$\text{So, final answer is, } \frac{-2}{1-sin2x}$$

Q No. 5:-

Answer(5):- we have given function, $f(x) = x^3 - 3x^2 + 2$.

So, differentiate this function, $f'(x) = 3x^2 - 6x$.

Take common, $f'(x) = 3x(x-2)$.

critical point occur where $f'(x) = 0$.

So, $3x(x-2) = 0$.

This gives us, $x = 0$, & $x = 2$.

- **If function is < 0 then its decreasing.**
- **If function is > 0 then its increasing.**

These points divide the x-axis into three intervals: $(-\infty, 0)$, $(0, 2)$ & $(2, \infty)$.

- **To check the function whether $f(x)$ is increasing or decreasing in each interval.**
 - (i). interval $(-\infty, 0)$:**

We consider, $f'(-1)$.

$$f(x) = 3x(x-2)$$

$$f(-1) = 3(-1)(-1-2)$$

$$f(-1) = -3(-3)$$

$f(-1) = 9$ so we saw that $f(-1) > 0$ so its function is increasing.

(ii). Interval $(0, 2)$:

We consider $f'(1)$.

$$f'(x) = 3x(x-2)$$

$$f'(1) = 3(1)(1-2)$$

$$f'(1)=3(-1)=-3$$

so we saw that $f'(1)<0$ so its function is decreasing.

(iii). Interval $(2,\infty)$

We consider, $f'(3)$.

$$f'(x)=3x(x-2).$$

$$f'(x)=3(3)(3-2)$$

$$f'(x)=9$$

so we saw that $f'(3)>0$ it's function is increasing.

- **Summary:**

- Increasing at $(-\infty,0) \cup (2,\infty)$.

- Decreasing at $(0,2)$.

Q No. 6:-

Answer(6):- (i). $\int \log x \, dx$

We consider formula in the form of (u & v).

So, $\int 1 \cdot \log x \, dx$

$$\text{Or, } \log x \int 1 \, dx - \int \frac{d(\log x)}{dx} \int 1 \, dx \, dx + c$$

$$\text{Or, } \log x \cdot x - \int \frac{1}{x} \cdot x \, dx$$

$$\text{Or, } x \log x - x + c.$$

So, final answer is: $x \log x - x + c$.

$$\text{(ii). } \int_0^1 \frac{x^3}{\sqrt{1-x^8}} \, dx$$

$$= \int_0^1 \frac{x^8}{\sqrt{1-(x^4)^2}} \, dx$$

Put $x^4=t$.

Diff. both side ;

$$4x^3 dx = dt$$

$$\text{Or, } x^3 dx = dt/4.$$

$$\text{Now, } \frac{1}{4} \int_0^1 \frac{dt}{\sqrt{1-t^2}}$$

$$\text{Or, } \frac{1}{4} \sin^{-1} t + c$$

$$\text{Or, } \frac{1}{4} \sin^{-1}(x^4) + c$$

Applying limit:

$$\int_0^1 \frac{x^3}{\sqrt{1-x^8}} dx = \frac{1}{4} \cdot \frac{\pi}{2} = \frac{\pi}{8}.$$

So, final answer is : $\frac{\pi}{8}$

➤ **Final answer of (i) & (ii) is**

- **The solution of $\int \log x dx$ is;**
= $x(\log x - 1) + c$
- **The final solution of $\int_0^1 \frac{x^3}{\sqrt{1-x^8}}$ is;**
= $\frac{\pi}{8}$.